

A Comparative Study of Image Inpainting Methods under Diverse Masking Conditions

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Abstract

In this work, we investigate and compare three representative deep learning architectures for the image inpainting task: Partial Convolutions, Contextual Attention, and the Iizuka model. All methods are evaluated using pre-trained models on a common test setting, focusing on their ability to handle different types of structured and unstructured corruption. In particular, we consider a diverse set of masks varying in shape, spatial distribution, and corruption ratio.

To assess performance, we employ both pixel-wise metrics (L_1 error, PSNR) and perceptual metrics (SSIM, LPIPS), allowing us to capture different aspects of reconstruction quality.

Our results show that no single method consistently dominates across all scenarios. Contextual Attention and the Iizuka model with blending generally achieve the best overall performance, while Partial Convolutions exhibit more limited reconstruction quality. Furthermore, the effectiveness of each method strongly depends on the type and size of the missing regions, highlighting the importance of mask diversity in evaluation.

1. Introduction

Image inpainting refers the process of filling in portions of images that are damaged, deteriorated, or missing. In particular, when doing non-blind inpainting, the location of the corrupted pixels is provided in order to fill them with plausible imagery. Therefore, it is relevant in many applications, such as editing out unwanted objects or generating occluded regions. Our work focuses on testing the performance of 3 different pre-trained networks on this task, highlighting how it varies based on the masked region shape.

2. Related Work

Image inpainting has been approached in many different ways: starting from deterministic algorithms relying on propagating the local image appearance around the target holes to fill them in. These diffusion-based approaches

can deal with small and narrow holes, while they struggle with large ones where they tend to result in oversmoothing. Patch-based methods, instead, rely on individuating relevant patches from the image’s non-hole regions iteratively in order to fill in the blanks. This procedure usually comes with a significant computational cost [10], which is alleviated by the randomized patch search algorithm Patch-Match [2]. However, since there is no learning in this approach, it is impossible to make semantically aware patch selections or to generate new objects not found in the original image. As for the deep learning approaches, we refer to the works our analysis is based upon: Partial Convolution [7], Contextual Attention [11] and Iizuka model [5].

3. Dataset

In order to test the selected approaches, we utilize a subset of the Places365 test set, which contains unique imagery related to 365 scene categories. In particular, due to computing resources constraints, we consider 200 images from the resized test set, where the images have been resized to $256 * 256$ regardless of the original aspect ratio. For each image, we consider its combination with each one of our 48 masks, totaling 9600 unique inputs to the networks. These masks are classified into 4 categories, according to their generation procedure: "salt and pepper", circular, box and random walk. We do not apply any global preprocessing to the selected images nor any augmentation, since each approach has its own peculiar procedure: this is, however, fairly simple in all approaches, consisting of a combination of resizing, central cropping and normalization using ImageNet statistics. We reckon this type of very light preprocessing is suited to the inpainting task, where it is important to preserve the pixel information from the ground truth. Moreover, not utilizing augmentation is also justifiable, since some operations, like random crop or color jitter, might actually be detrimental for this specific task. In fact, they risk to, respectively, restrict the input too close to the mask (therefore losing important context) or blur the boundary around the inpainted section.

4. Models

4.1. Partial Convolution

For the approach involving partial convolutions we utilize a UNet-like architecture, replacing all convolutional layers with partial convolutional layers and using nearest neighbor up-sampling in the decoding stage. One of the novelties of this method is that convolutional results depend only on the non-hole regions at every layer and it is possible to introduce a mask update step that can remove any masking for those pixels where the partial convolution has been able to operate on an unmasked value, gradually shrinking the masked region [7]. Formally, the *Partial Convolutional Layer* is defined as

$$x' = \begin{cases} \mathbf{W}^T (\mathbf{X} \odot \mathbf{M}) \frac{\text{sum}(\mathbf{1})}{\text{sum}(\mathbf{M})} + b, & \text{if } \text{sum}(\mathbf{M}) > 0 \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

where $\mathbf{W}$ is the convolution filter weights, b its corresponding bias, $\mathbf{X}$ the feature values for the current convolution window and $\mathbf{M}$ is the corresponding binary mask. The symbol $\odot$ denotes element-wise multiplication, and $\mathbf{1}$ has same shape as M but with all elements being 1. For the mask update, if the convolution is able to condition its output on at least one valid input value, then that location is marked as valid. As far as the loss function is concerned, it consists of multiple terms: $\mathcal{L}_{hole}$ and $\mathcal{L}_{valid}$, which are the L_1 losses on the network output for the hole and the non-hole pixels respectively. These two terms, however, do not measure how well the generated portion blends with the non masked region, so we consider the additional terms $\mathcal{L}_{perceptual}$, $\mathcal{L}_{style}$ and $\mathcal{L}_{tv}$. The perceptual loss, introduced by Gatys et al. [4] is defined as:

$$\mathcal{L}_{perceptual} = \sum_{p=0}^{P-1} \frac{\|\Psi_p^{\mathbf{I}_{out}} - \Psi_p^{\mathbf{I}_{gt}}\|_1}{N_{\Psi_p^{\mathbf{I}_{gt}}}} + \sum_{p=0}^{P-1} \frac{\|\Psi_p^{\mathbf{I}_{comp}} - \Psi_p^{\mathbf{I}_{gt}}\|_1}{N_{\Psi_p^{\mathbf{I}_{gt}}}} \quad (2)$$

where $\mathbf{I}_{comp}$ is the raw output image $\mathbf{I}_{out}$, but with the non-hole pixels directly set to ground truth; $N_{\Psi_p^{\mathbf{I}_{gt}}}$ is the number of elements in $\Psi_p^{\mathbf{I}_{gt}}$. The style-loss term is similar to the perceptual loss [4], but we first perform an autocorrelation (Gram matrix) on each feature map before applying the L_1 . Finally, the total variation (TV) loss represents the smoothing penalty [6] on R , where R is the region of 1-pixel dilation of the hole region. The total loss $\mathcal{L}_{total}$ is the combination of all the above loss functions.

$$\mathcal{L}_{total} = \mathcal{L}_{valid} + 6\mathcal{L}_{hole} + 0.05\mathcal{L}_{perceptual} + 120(\mathcal{L}_{style_{out}} + \mathcal{L}_{style_{comp}}) + 0.1\mathcal{L}_{tv} \quad (3)$$

The loss term weights are taken from the original work [7], where they were determined by performing a hyperparameter search on 100 validation images. The pre-trained model we use was trained on the Places dataset for 500000 iterations [8].

4.2. Iizuka model

The model proposed by Iizuka et al.[5] addresses the image inpainting problem by combining a completion network with adversarial training based on both global and local context constraints. The architecture builds upon the Context Encoder framework, extending it with a more structured adversarial setup. In particular, the model is composed of three main components shown in Fig.1: a completion network and two discriminators, namely a global context discriminator and a local context discriminator. The completion network is a fully convolutional encoder-decoder architecture designed to fill missing regions while preserving the known pixels unchanged. The encoder reduces the spatial resolution using strided convolutions limiting the use of pooling layers to avoid excessive blurring, while the decoder restores the resolution through deconvolution layers. The discriminators play a key role during training: the global discriminator processes the entire image to enforce overall semantic consistency, while the local discriminator focuses on a small region centered around the hole to ensure fine-grained realism. Their outputs are feature vectors that are concatenated and mapped, through a fully connected layer with sigmoid activation, to a probability value indicating whether the image is real or completed. Training is performed by alternating optimization steps: first the discriminators are trained to distinguish real images from completed ones, and then the completion network is updated to fool them, following an adversarial objective. The overall loss combines a reconstruction term and a GAN term. The reconstruction loss is a weighted Mean Squared Error defined as:

$$L(x, M_c) = \|M_c \odot (C(x, M_c) - x)\|_2^2, \quad (4)$$

where x is the ground truth image, M_c is the binary mask, $C(x, M_c)$ is the network output, and $\odot$ denotes element-wise multiplication. The adversarial objective is given by:

$$\min_C \max_D \mathbb{E}[\log D(x, M_d) + \log(1 - D(C(x, M_c), M_c))] \quad (5)$$

where D represents the joint discriminator, and the expectation is taken over the training distribution. Training is divided into three phases: pretraining of the completion network with the MSE loss, separate training of the discriminators, and finally joint adversarial training. Training masks are generated as random rectangular holes filled with the dataset mean value. The model may still suffer from slight color inconsistencies between generated and known regions; for this reason, a post-processing step based on fast marching and Poisson image blending [9] is applied to improve boundary coherence. We utilize pre-trained weights [1] for the model, which was originally trained on the Places2 dataset containing over eight million images.

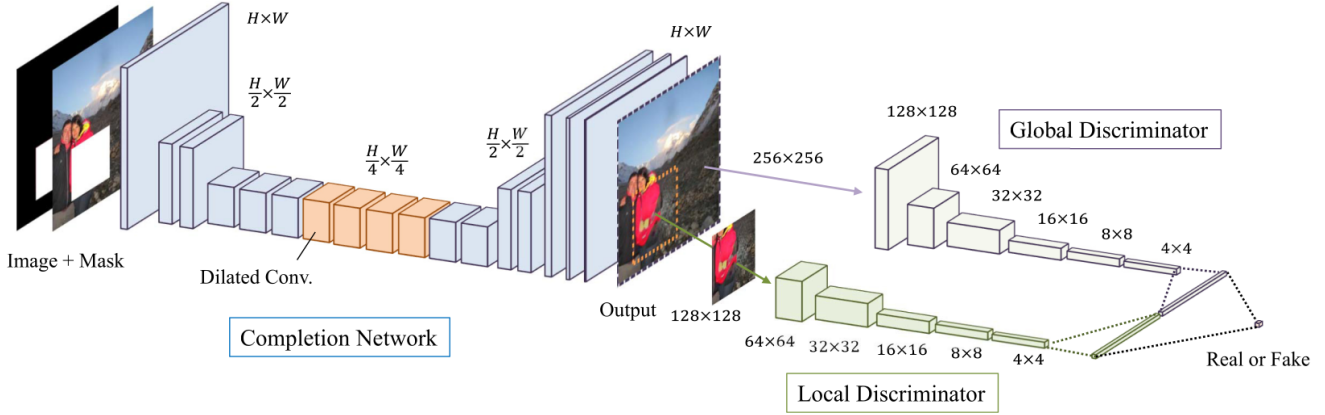


Figure 1. Architecture of the Iizuka et al. model.

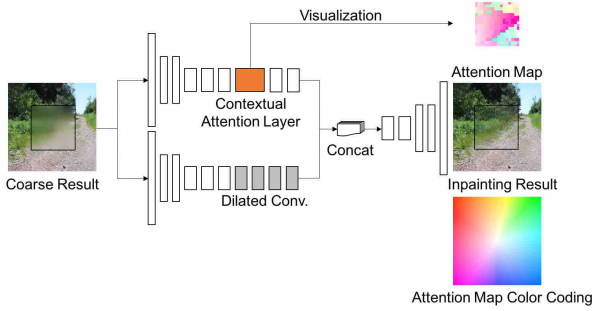


Figure 2. Based on coarse result from the first encoder-decoder network, two parallel encoders are introduced and then merged to single decoder to get in-painting result. For visualization of the attention map, white (center of color coding map) means the pixel attends on itself, pink on bottom-left, green means on top-right [11].

4.3. Contextual Attention

The architecture chosen by the authors [11] for this approach starts with a two-stage coarse-to-fine network architecture where the first network makes an initial coarse prediction, and the second network takes the coarse prediction as inputs and predicts refined results. The coarse network is trained with the reconstruction loss explicitly, while the refinement network is trained with the reconstruction as well as GAN losses. Interestingly, no batch normalization layers are used and, as activation function, ELU is utilized. A clever addition to the loss definition is the use of a spatially discounted reconstruction term using a weight mask, due to the fact that pixels near the mask boundaries have much less ambiguity than those pixels closer to the center of the hole. Thus, the weight of each pixel in the mask is computed as γ^l , where l is the distance of the pixel to the nearest known pixel and γ is set to 0.99 in all experiments [11]. Therefore, to integrate the attention module, two parallel encoders are

introduced as shown in Fig.2. The bottom encoder specifically focuses on hallucinating contents with layer-by-layer (dilated) convolution, while the top one tries to attend on background features of interest. Output features from the two encoders are, finally, aggregated and fed into a single decoder to obtain the output. The pre-trained model we use was trained on ImageNet for 430000 iterations [3].

5. Experiments

The focus of our work lies on studying the performance of different approaches to the in-painting task with respect to the type of mask used. We evaluate the models using different types of masks in order to simulate a wide range of realistic and controlled corruption patterns. In particular, we consider SAP (salt and pepper), box, circle, and random walk masks, each designed to stress different aspects of the in-painting task. SAP masks introduce sparse and unstructured pixel corruption, mainly testing local texture reconstruction and denoising capabilities. Box masks remove large contiguous regions and are used to evaluate the ability of the model to recover missing semantic content and global structure. Circle masks have a similar missing area to box masks but with smoother boundaries, allowing us to study the influence of boundary geometry on reconstruction quality. Random walk masks generate irregular missing regions, which better resemble real-world artifacts such as scratches or object removal and require stronger long-range contextual reasoning. We utilize the corruption ratio $\{0.10, 0.25, 0.40\}$ to analyze how performance degrades as the missing area increases. Additionally, we generate masks located along both the periphery and the central region of the image.

For these reasons, and due to computational and time constraints, we construct a total of 48 masks designed to cover a broad range of possible corruption scenarios by combining different generation methods, corruption ratios,

and degrees of proximity to image borders. To systematically analyze model behavior, each mask is applied to every image in the test set. The complete set of masks used in our experiments is reported in Fig. 3.

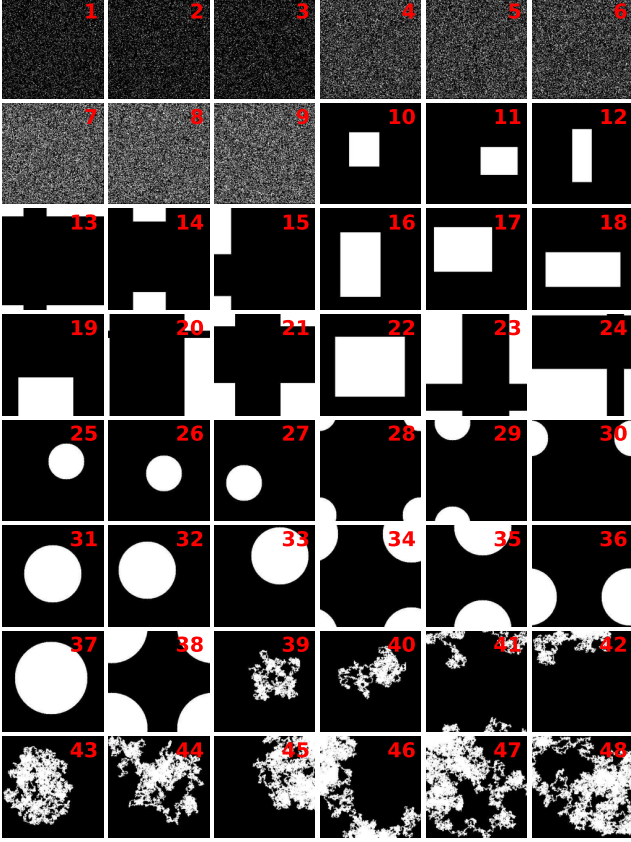


Figure 3. Masked region in white. In the top right of each mask there is its index designation: In order sap, box, circle and random walk. For each type, images with masked pixels to total pixels ratio in $\{0.10, 0.25, 0.40\}$ are generated, first not allowing to cross image borders and then permitting it.

5.1. Metrics

To evaluate the quality of the reconstructed images, we use both pixel-wise and perceptual metrics to capture different aspects of similarity between the predicted image $\hat{I}$ and the ground truth image I .

The simplest metric is the L_1 error, defined as the average absolute difference over all pixels. While this metric is stable, easy to compute, and directly interpretable (lower is better), it is overly sensitive to small spatial misalignments.

A closely related metric is PSNR (Peak Signal-to-Noise Ratio), which is based on the mean squared error (MSE):

$$PSNR = 10 \log_{10} \left(\frac{MAX^2}{MSE} \right) \quad (6)$$

Here MAX is the maximum possible pixel value. PSNR can be interpreted as a signal-to-noise ratio in decibels, so

the higher the better. Compared to L_1 , it penalizes large errors more strongly due to the squared term.

To better match human perception, we also use SSIM (Structural Similarity Index), which compares images in terms of luminance, contrast, and structure. For two image patches x and y , SSIM is defined as:

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (7)$$

Unlike L_1 and PSNR, SSIM operates on local windows and focuses on structural information such as edges and textures, making it more correlated with perceived visual quality. However, it still depends on a reference image.

Finally, we consider LPIPS (Learned Perceptual Image Patch Similarity), which measures distance in the feature space of a pretrained neural network (in our case, AlexNet). Given features $\phi_l(x)$ from layer l , the metric is:

$$LPIPS(x, x_0) = \sum_l w_l \frac{1}{H_l W_l} \sum_{h,w} \|\hat{\phi}_l(x)_{h,w} - \hat{\phi}_l(x_0)_{h,w}\|_2^2 \quad (8)$$

Here $\hat{\phi}_l$ denotes normalized features and w_l are learned weights. LPIPS captures perceptual differences better than pixel-wise metrics, since it compares high-level representations rather than raw intensity values. Nevertheless, it still requires a ground-truth reference and depends on the choice of the pretrained network.

Since image inpainting is inherently a one-to-many problem, multiple visually plausible solutions may exist for a given missing region. All the metrics above implicitly assume the existence of a single correct ground truth. As a result, different plausible completions can obtain worse quantitative scores even if they are perceptually realistic and semantically correct.

5.2. General results

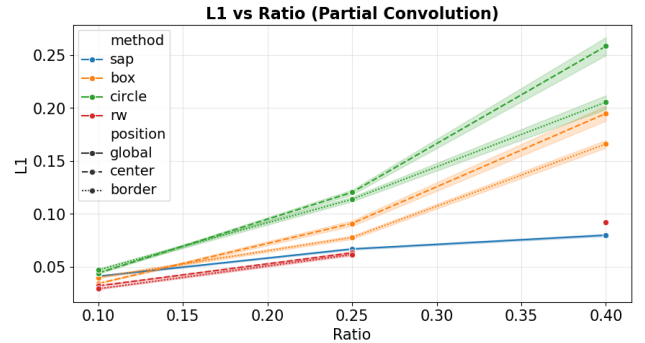


Figure 4. Metrics in function of the masked pixels ratio. Due to space limitations only Partial Convolution with L_1 is shown, however the other models and metrics behave in the same way.

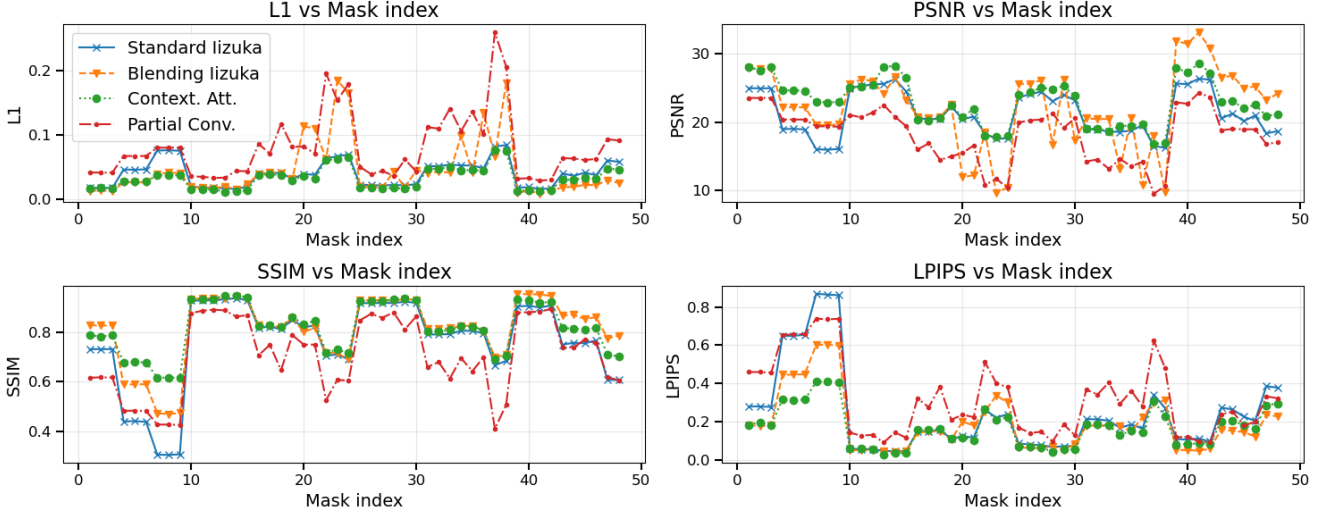


Figure 5. Model comparison over the different masks: the trend is similar across all models for every metric.

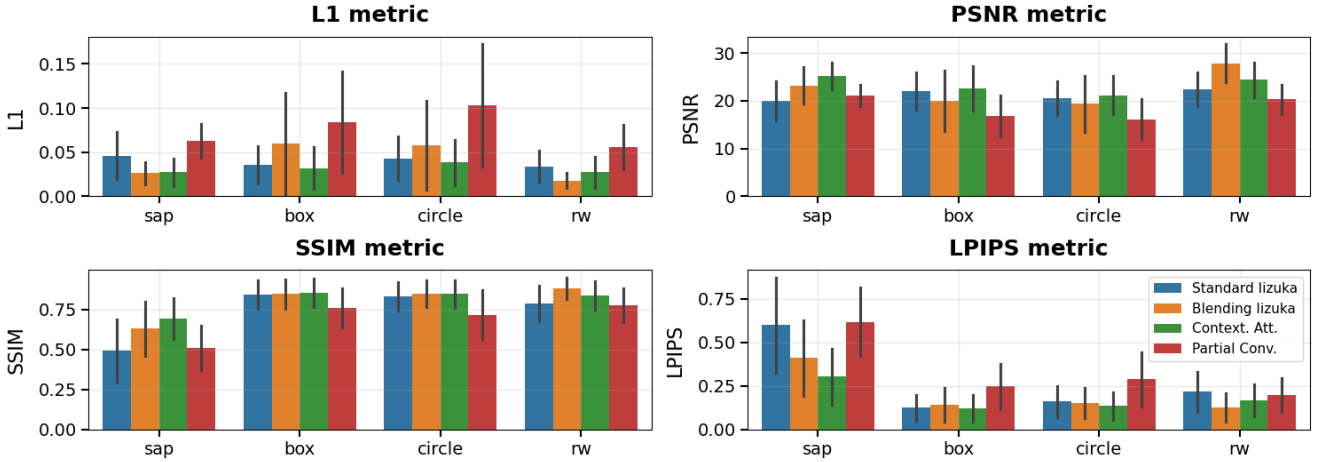


Figure 6. Model comparison over the different mask shapes

First of all, as we can see from Fig.4, increasing the corrupted portion of the image makes for a more challenging tasks and all the metrics worsen with respect to smaller masks. Moreover, up to 0.25 corruption ratio, there is not much difference between generating masks near or far from the border. Instead, when the masked portion ratio increases, the "border" generation yields better results. This can be explained because with the former setting, the mask is more likely to cover the image in its borders (since it can loop), where there is less semantically important information, which is, instead, mostly present in the center of the image. Indeed, it is something that all approaches struggle with, as can be seen in Fig.5, where it is evident that all models share the same "difficult" (poor performing) mask shapes. However, the relative ranking of the networks is not the same across mask types (Fig.6): for simple and regular masks (e.g. box and circle with small or medium

ratios), all models except Partial Convolution show comparable behavior, whereas for more scattered masks (SAP and random walk) clearer differences emerge, with Contextual Attention performing slightly better on sparse noise and Iizuka with blending showing stronger robustness on irregular structures. We report in Table 1 the mean values of the metrics for each model.

Table 1. Mean values for each of the considered metrics

Model	L1	PSNR	SSIM	LPIPS
Standard Iizuka	0.039	21.3	0.761	0.243
Blending Iizuka	0.044	22.0	0.813	0.192
Context. Att.	0.032	23.0	0.817	0.169
Partial Conv.	0.079	18.2	0.703	0.318

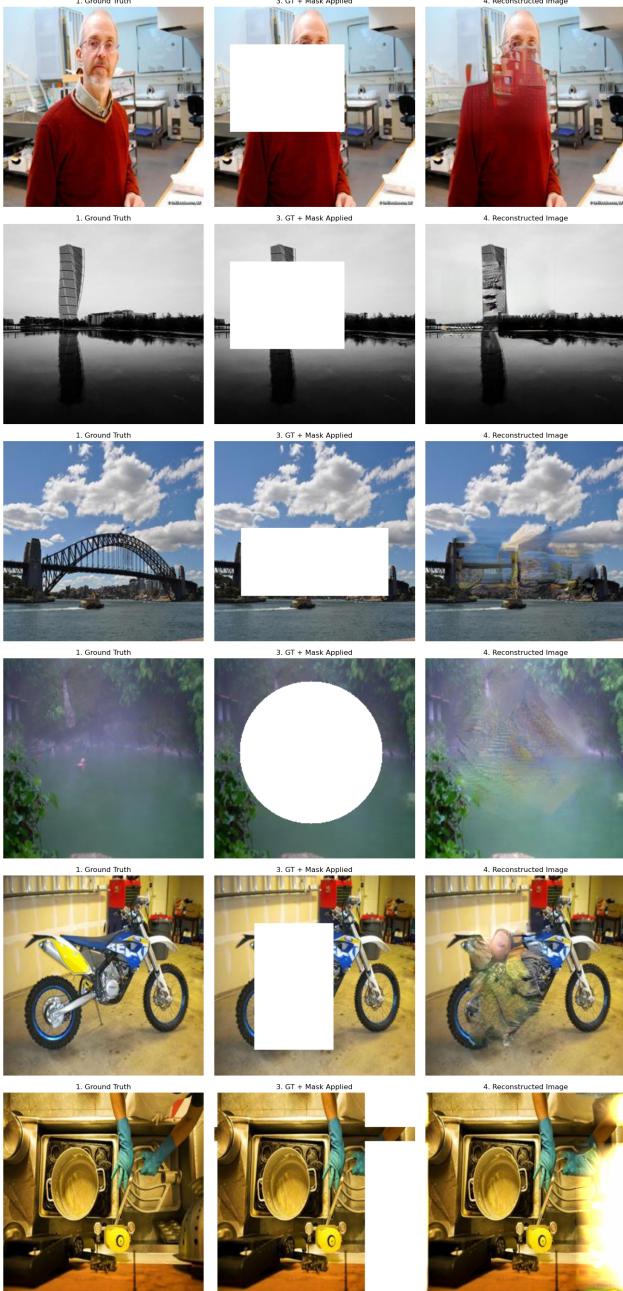


Figure 7. Some Iizuka model results. Image index vertically ordered from *a* to *f*.

5.3. Iizuka model and the Poisson blending

Overall, the Iizuka model produces visually plausible completions, but its behavior strongly depends on the mask type and the amount of missing content. It generally performs a little better on structured geometric masks such as boxes compared to circles, possibly due to the type of regions seen during training. The model struggles when large semantic information is missing, often failing to recon-

struct coherent objects when only partial context is available (e.g., Fig. 7*a*). On the other hand, it can sometimes successfully hallucinate plausible object structures when enough contextual cues remain, such as partially visible buildings (e.g., Fig. 7*b*). Several typical failure modes are observed: color inconsistencies already in the base model output, unintended object elongation, and incorrect object removal or creation when masks overlap semantic entities (e.g., Fig. 7*c*). Performance is generally better when the surrounding scene is visually uniform (e.g., Fig. 7*d*), while structured patterns make artifacts more visible to the human eye (e.g., Fig. 7*e*).

The addition of Poisson blending generally improves visual consistency by correcting color shifts and smoothing transitions between generated and known regions. In SAP masks, blending tends to reduce noise artifacts. For bordered masks, blending can be harmful when the missing region is large, as the lack of sufficient boundary constraints in the Poisson formulation can lead to unstable solutions, sometimes producing over-saturated or nearly white patches (e.g., Fig. 7*f*). In centered masks, blending is beneficial for small missing regions, and becomes more effective as the patch size increases and color inconsistencies become stronger. For random walk masks, blending helps in correcting global color distribution and hiding irregular artifacts.

6. Conclusion

In this work, we evaluated several image inpainting approaches, namely Partial Convolution, Contextual Attention, and the Iizuka model (with and without Poisson blending), using a diverse and controlled set of masks. The analysis highlights how the performance of inpainting models strongly depends on both the structure and the size of the missing regions, confirming that the task is inherently complex and highly context-dependent.

From a quantitative perspective, Partial Convolution consistently achieved the worst results across all considered metrics (L_1 , PSNR, SSIM, LPIPS), indicating limited ability to handle complex or irregular corruptions. In contrast, Contextual Attention and the Iizuka model with blending generally obtained the best scores, with similar overall performance.

More generally, both qualitative and quantitative results confirmed a fundamental limitation of standard evaluation metrics for inpainting, due to the task admitting multiple plausible solutions.

Future studies could use a larger dataset of images and masks, which was limited here by time and computing power. Moreover, in order to address the one-to-many problem inherent to image inpainting, it could be beneficial to incorporate human evaluation to judge the quality of different outputs.

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